

Convolutional Neural Networks



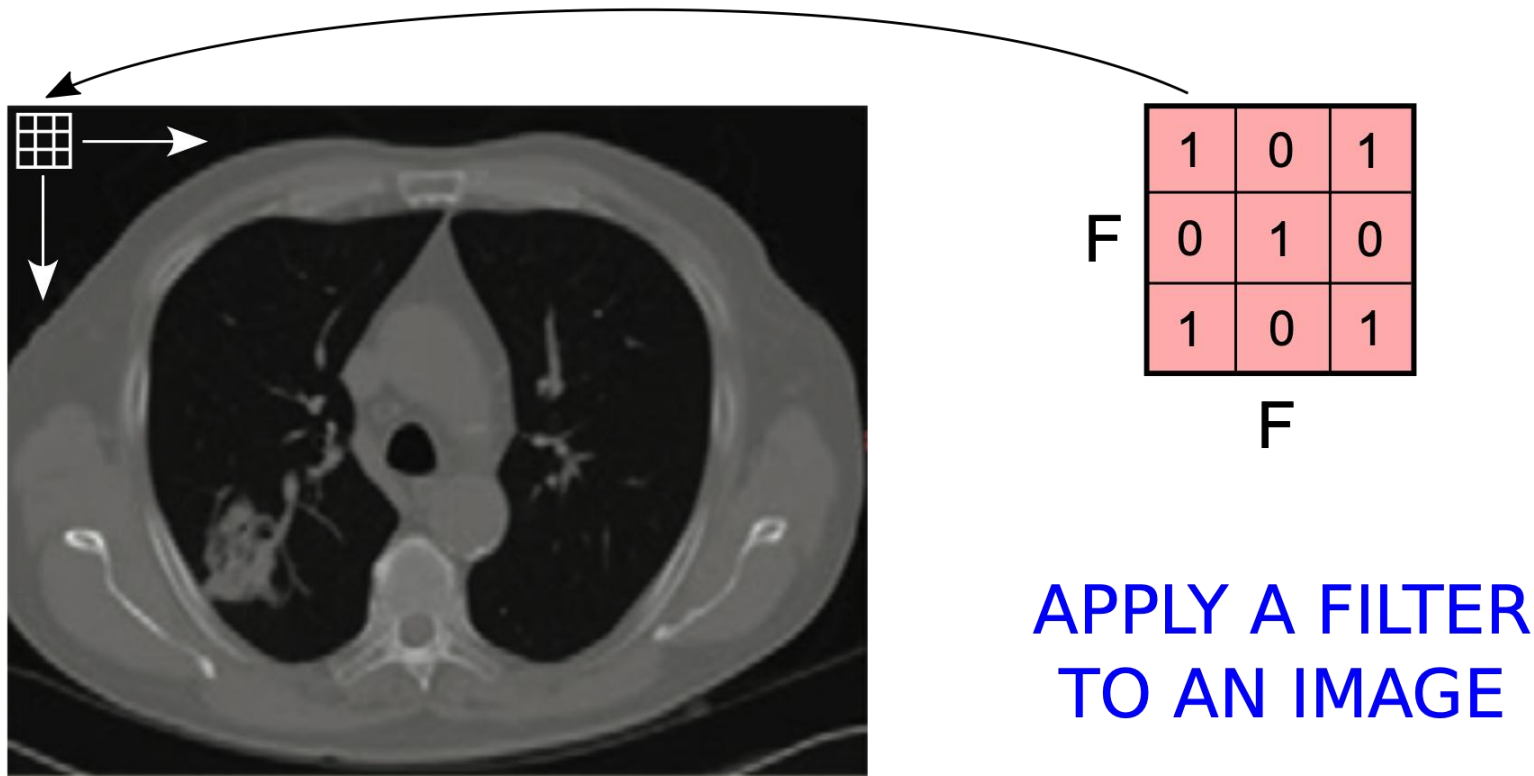
Convolutional Neural Networks



One of the most **widely** used deep network models

- First idea **dates back** to the 80s-90s!!!
 - Initially designed to classify **images**
 - Later on extended to **other** inputs (e.g., text)
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- I could also apply MLP to images, but I'd lose geometry information

Convolutional Neural Networks



Convolutional Neural Networks

1 _{x1}	1 _{x0}	1 _{x1}	0	0
0 _{x0}	1 _{x1}	1 _{x0}	1	0
0 _{x1}	0 _{x0}	1 _{x1}	1	1
0	0	1	1	0
0	1	1	0	0

Image

4		

Convolved
Feature

Convolutional Neural Networks

1	1 _{x1}	1 _{x0}	0 _{x1}	0
0	1 _{x0}	1 _{x1}	1 _{x0}	0
0	0 _{x1}	1 _{x0}	1 _{x1}	1
0	0	1	1	0
0	1	1	0	0

Image

4	3	

Convolved
Feature

Convolutional Neural Networks

1	1	1 _{x1}	0 _{x0}	0 _{x1}
0	1	1 _{x0}	1 _{x1}	0 _{x0}
0	0	1 _{x1}	1 _{x0}	1 _{x1}
0	0	1	1	0
0	1	1	0	0

Image

4	3	4

Convolved
Feature

Convolutional Neural Networks

1	1	1	0	0
0 _{x1}	1 _{x0}	1 _{x1}	1	0
0 _{x0}	0 _{x1}	1 _{x0}	1	1
0 _{x1}	0 _{x0}	1 _{x1}	1	0
0	1	1	0	0

Image

4	3	4
2		

Convolved
Feature

Convolutional Neural Networks

1	1	1	0	0
0	1 _{x1}	1 _{x0}	1 _{x1}	0
0	0 _{x0}	1 _{x1}	1 _{x0}	1
0	0 _{x1}	1 _{x0}	1 _{x1}	0
0	1	1	0	0

Image

4	3	4
2	4	

Convolved
Feature

Convolutional Neural Networks

1	1	1	0	0
0	1	1 _{x1}	1 _{x0}	0 _{x1}
0	0	1 _{x0}	1 _{x1}	1 _{x0}
0	0	1 _{x1}	1 _{x0}	0 _{x1}
0	1	1	0	0

Image

4	3	4
2	4	3

Convolved
Feature

Convolutional Neural Networks

1	1	1	0	0
0	1	1	1	0
0 _{x1}	0 _{x0}	1 _{x1}	1	1
0 _{x0}	0 _{x1}	1 _{x0}	1	0
0 _{x1}	1 _{x0}	1 _{x1}	0	0

Image

4	3	4
2	4	3
2		

Convolved
Feature

Convolutional Neural Networks

1	1	1	0	0
0	1	1	1	0
0	0 _{x1}	1 _{x0}	1 _{x1}	1
0	0 _{x0}	1 _{x1}	1 _{x0}	0
0	1 _{x1}	1 _{x0}	0 _{x1}	0

Image

4	3	4
2	4	3
2	3	

Convolved
Feature

Convolutional Neural Networks

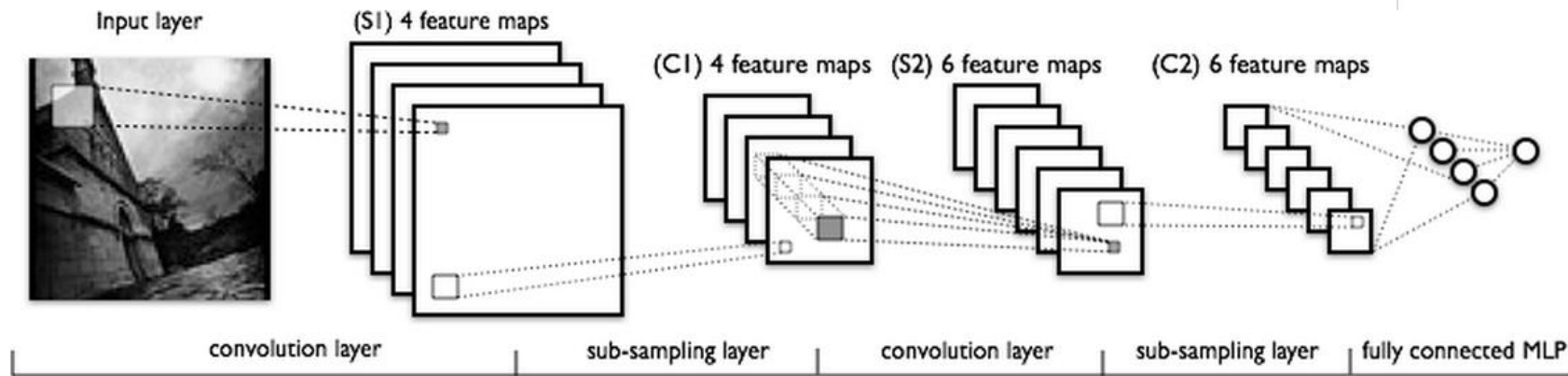
1	1	1	0	0
0	1	1	1	0
0	0	1 _{x1}	1 _{x0}	1 _{x1}
0	0	1 _{x0}	1 _{x1}	0 _{x0}
0	1	1 _{x1}	0 _{x0}	0 _{x1}

Image

4	3	4
2	4	3
2	3	4

Convolved
Feature

Convolutional Neural Networks



CNNs are **stacks** of (convolution) layers

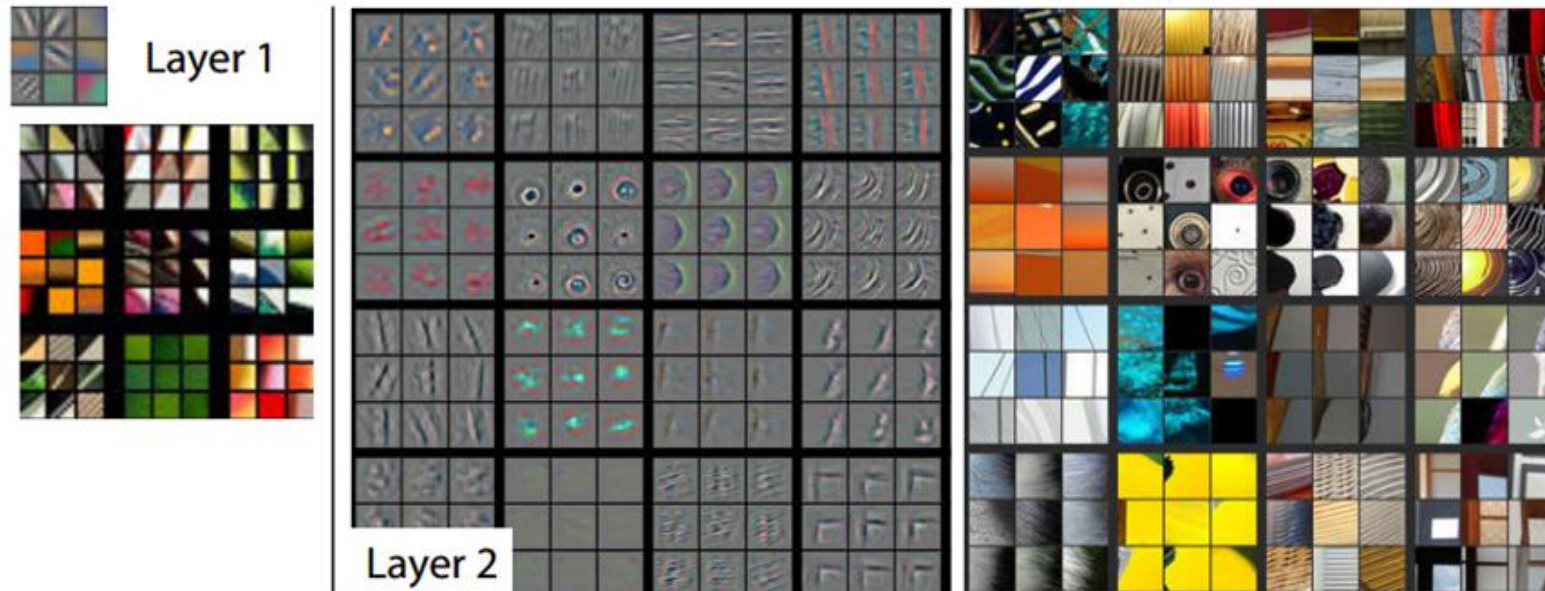
Automatic **hierarchical** feature extraction

Learning a **representation** (features) of the input

Filters are **learned** from data

Convolutional Neural Networks

Examples of learned filters



Visualizations of Layer 1 and 2. Each layer illustrates 2 pictures, one which shows the filters themselves and one that shows what part of the image are most strongly activated by the given filter. For example, in the space labeled Layer 2, we have representations of the 16 different filters (on the left)

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